


There are different types of fermentor used in industrially micro biology which includes.

1. Stirred tank bioreactor
2. Tower bioreactors
3. Air lift bioreactors
4. Packed–bed bioreactors
5. Fluidized bed bioreactors
6. Photo bioreactors

### 6.5.2 Media Used in the Industrial Productions

**Fermentation Medium**

Most fermentation requires liquid media, often referred to as broth, although some solid-substrate fermentations are also operated. Fermentation media must satisfy all the nutritional requirements of the microorganism and fulfill the technical objectives of the process. Animal fats and plant oils are also incorporated into some media, often as supplements to the main carbon source.

Medium used for large scale production should have the following characteristics.

1. It should be cheap and easily available.
2. It should maximize the growth of the microorganism productivity and the rate of formation of the desired product.
3. It should minimize the formation of undesired products.

It should contain carbon source, nitrogen source, energy source, micro nutrients required for the industrial production. Table 6.3 shows common substances used in the industrial fermentation process.

Waste products from other industrial processes such as molasses, ligno cellulosic waste, and corn steep liquor are generally used as substrates for industrial fermentation.

Apart from carbon and nitrogen sources, some other components like minerals, vitamins, growth factors are also used in Industrial fermentations.

**Minerals**

Normally, sufficient quantities of cobalt, copper, iron, manganese, molybdenum, and zinc are present in the water supplies, and as impurities in other media ingredients. For example, corn steep liquor contains a wide range of minerals that will usually satisfy the minor and trace mineral needs.

**Vitamins and growth factors**

Many bacteria can synthesize all necessary vitamins from basic elements. For other bacteria, filamentous fungi and yeasts, they must be added as supplements to the fermentation medium. Most natural carbon and nitrogen sources also contain at least some of the required vitamins as minor contaminants. Other necessary growth factors, amino acids, nucleotides, fatty acids and sterols, are added either in pure form or, for economic reasons, as less expensive plant and animal extracts.

**Precursors**

Some fermentation must be supplemented with specific precursors, notably for secondary metabolite production. When required, they are often added in controlled quantities and in a relatively



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